

The principle of the injection laser diode as shown in figure 16-3. The natural light emitted due to the recombination of the electrons and positive holes is caught between two mirror plates; so photons can excite other photons, by injecting their light energy in the PN junction, thus creating more electrons and positive holes etc.

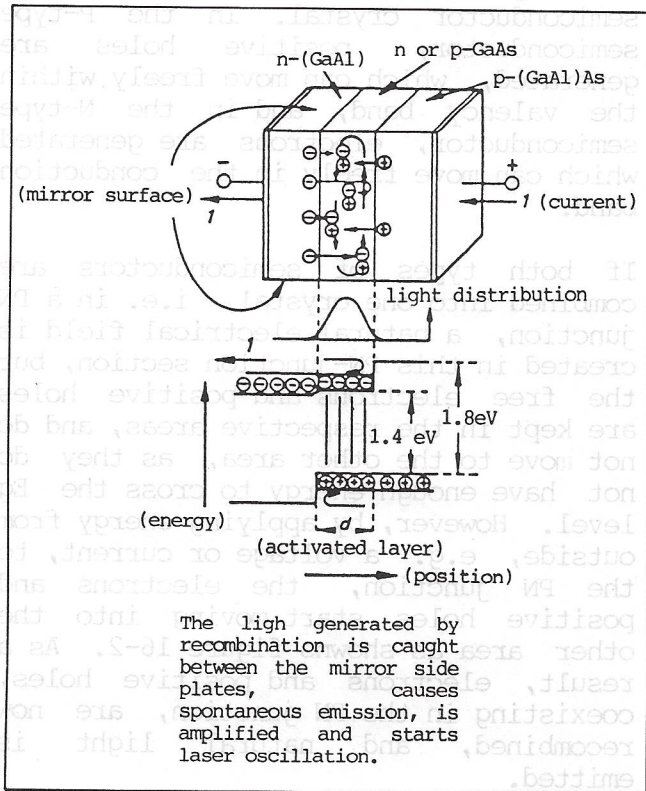


Figure 16-3 As Double Hetero Laser Diode

The light generated by recombination is caught between the mirror side plates, causes spontaneous emission, is amplified and starts laser oscillation. This stimulated emission can be so rapid as to represent nearly an avalanche, completely draining the high-energy states in a subnanosecond burst of very great energy.

On the other hand, certain types of lasers can be produced which will operate in an equilibrium condition, giving up photons at just the input energy pumping rate. The injection laser diode operates in this mode.

The material in an injection laser diode is rather heavily doped so that, under forward bias, the region near the junction has a very high concentration of holes and electrons. This produces the inversion condition with a large population of electrons in a high-energy band and a large population of holes in the low-energy band.

In this condition, the stimulated emission of photons can overcome the photon absorption and a net light flux will result. In operation, the forward current must reach some threshold, to achieve lasing operation. Just beyond this threshold, the laser operates with a single "thread" of light, and the output is relatively stable but low.

As the current is increased, the light output climbs extremely rapidly. The light output is highly temperature sensitive. A small current variation or modest temperature change can cause the output to rise so rapidly that it destroys the chip.

A photodiode, monitoring the light output, is commonly used in a feedback loop to overcome this problem. Like the LED, the ILD must be driven from a current source, rather than a voltage source as the unit can go into thermal runaway.

As a device begins to draw current, the junction begins to warm and forward drop decreases, which tends to drive the voltage higher and in turn decrease the forward drop until the current tends toward infinity and the device is destroyed.

In addition, the ILD has a tendency to deterioration with operation, and this deterioration is greatly accelerated when operating the ILD outside of its optimum limits. Therefore, in an optical pickup assembly the operating point set at the factory should not be touched without the proper measuring equipment.

16. Optical Pickup

1. Basic Operation of the Laser Diode

The basic operation of any laser consists of pumping the atoms into a stimulated state from which the electrons can escape and fall to the lower energy ground state by giving up a photon of the appropriate energy. In a solid-state laser such as the laser diodes used in CD player, this principle is applied to a PN-junction, as will be explained below.

First, let us review the energy band diagrams as shown in figure 16-1.

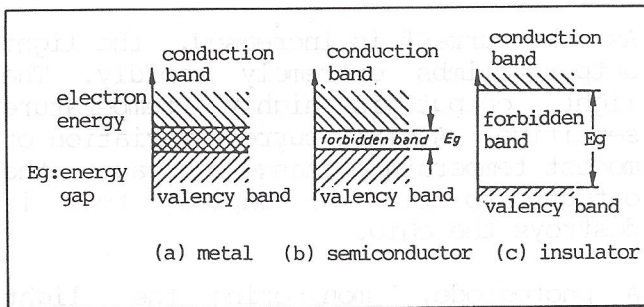


Figure 16-1 Energy Band

As is commonly known, the energy state of electrons falls within a certain range or energy band.

The energy bands required for electrical conductivity are:

(a) The "valence band", indicating the energy level of the valence electron and:

(b) The "conduction band", indicating the energy level of the free electron.

In a metal, the valence band and conduction band overlap each other in part, so that the valence electron can enter the conduction band and move freely, which causes conductivity.

In a semiconductor or insulator, there is an energy gap between the valence band and the conduction band; this band is called the forbidden band. In case of an insulator, the forbidden or energy gap " E_g " is large.

For a semiconductor, the energy gap is rather small, so that it can become a conductor by applying some external energy.

P-type and N-type semiconductors can be made by inserting impurities into the semiconductor crystal. In the P-type semiconductor, positive holes are generated, which can move freely within the valence band, and in the N-type semiconductor, electrons are generated which can move freely in the conduction band.

If both types of semiconductors are combined into one crystal, i.e. in a PN junction, a natural electrical field is created in this PN-junction section, but the free electrons and positive holes are kept in the respective areas, and do not move to the other area, as they do not have enough energy to cross the E_g level. However, by applying energy from outside, e.g. a voltage or current, to the PN junction, the electrons and positive holes start moving into the other area as shown in figure 16-2. As a result, electrons and positive holes, coexisting in the PN junction, are now recombined, and natural light is emitted.

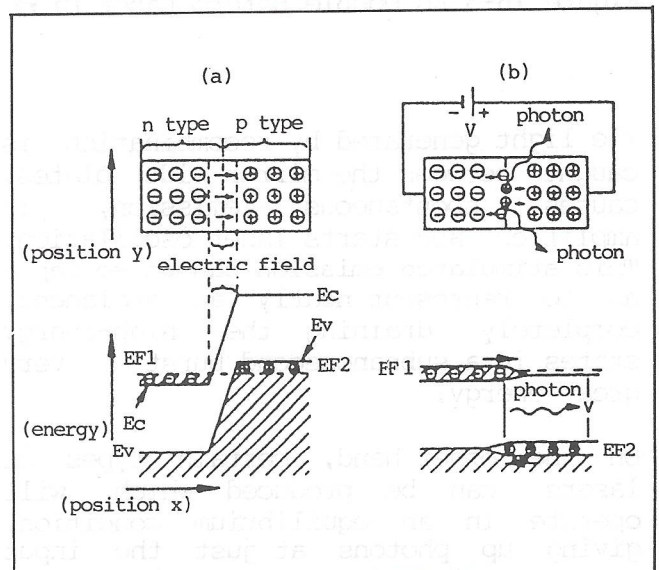


Figure 16-2 Light Emission

15) Description of Modified Servo Circuit (MN8373S and MN8374S)

1. General

The servo circuit in CD player with the standard mechanism (SL-P150, SL-P200, SL-P250, etc) has been entirely modified during production. The units containing modified servo circuit can be identified as follows:

- (1). Serial number suffix: Refer to supplement.
- (2). A caution label sticking: THE ORIGINAL DESIGN HAS BEEN CHANGED.

2. Circuit Configuration

The servo circuit used in the conventional FF-1 CD players comprised a servo IC (AN8370S) and discrete circuit. In contrast, the modified servo circuit consists of three IC chips with on discrete circuit.

- (1). IC-101 (servo amplifier)
=Processes analog servo signals.
- (2). IC-102 (servo controller)
=Processes digital servo signals.
- (3). IC-103 (servo driver)
=Drives the focus, tracking, and traverse coils.

The block diagram of IC-101 (AN8373S) and IC-102 (AN8374S) are shown in figure 17-33.

3. Departures from the Conventional Servo Circuit and some Cautions

While the new servo circuit replaces discrete components integrated circuits, the basic circuit configurations and operations are almost identical to those of the conventional servo circuit, with the exception of the following details:

- (1). PLL adjustment eliminated
=Servo signals are automatically pulled into the PLL lock range by using the CLK (88.2KHz) signal.
- (2). 2.5V reference for tracking error and focus error signals
=Because the new circuit operates on a signal +5V supply, the reference voltage for the error signals has been changed to 2.5V as shown in figure 17-32.

NOTE

The 2.5V reference is also used for turntable height adjustment.

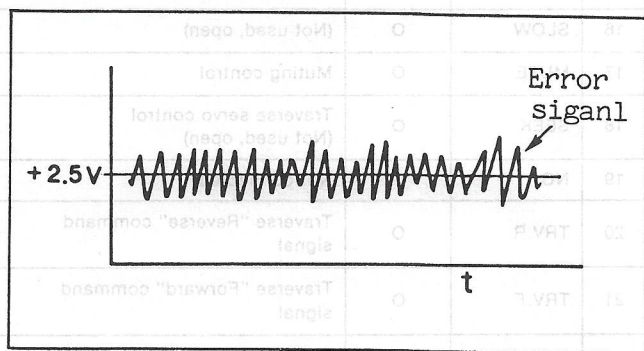


Figure 17-32

Focus and Tracking Error Signal

- (3). RF signal polarity inverted
=An inverted RF signal appears at J101. As a result, the RF and dropout waveforms have a polarity opposite to that of the conventional, as shown in figure 17-34.
- (4). Servo gain circuit and adjustment
=Focus: 825 Hz 150 mV
=Tracking: 1.2 KHz 150 mV
=Tracking balance: 1.1 KHz 600 mV

These voltages are input voltages required for adjustment.

14) Pin Description of MN1554PEP

● IC401: MN1554PEP (System Control)

Pin No.	Mark	I/O Division	Function
1	BRECV	—	(Not used, open)
2	BSEND	—	(Not used, open)
3	SYNC	O	(Not used, open)
4	SIRQ	I	Not used (connected to +5V)
5	BLKCK	I	Sub-code block (Q data) clock input (75Hz)
6	CLDCK	I	Sub-code block (Q data) clock input (7.35kHz)
7	SBO	I	(Not used, open)
8	SUBQ	I	Sub-code (Q data) input
9	RST	I	Reset signal input
10 } 13	P20 } P23	O	Not used (connected to +5V)
14	CLOSE	O	Loading motor "Close" command
15	OPEN	O	Loading motor "Open" command
16	SLOW	O	(Not used, open)
17	MUTE	O	Muting control
18	SEEK	O	Traverse servo control (Not used, open)
19	NC	—	Not connected
20	TRV.R	O	Traverse "Reverse" command signal
21	TRV.F	O	Traverse "Forward" command signal
22	CNT4	O	Optical servo IC control signal (KICKR: Kick direction [reverse] command)
23	CNT3	O	Optical servo IC control signal (KICKF: Kick direction [forward] command)
24	CNT2	O	Optical servo IC control (TRON: Tracking servo)
25	VDD	I	Power supply (connected to +5V)
26	DOWN	O	(Not used, open)
27	UP	O	(Not used, open)
28	CNT1	O	Optical servo IC control signal (FOON: Focus servo)

Pin No.	Mark	I/O Division	Function
29	CLOSE	I	Disc holder "Open" detection
30	OPEN	I	Disc holder "Close" detection
31	BCLK	I	(Not used, connected to GND)
32	BDATA	I	(Not used, connected to GND)
33	STAT	I	Processing status input from signal processing LSI
34	COMP	O	TOC reading control (ON at "L") (connected to GND)
35	FLOCK	I	Optical servo condition (focus) input
38	SENSE	I	Optical servo condition (track cross) input
37	RECV	I	Data receipt command signal
38	SEND	I	Data transmission command
39	ACK	I	Data discrimination signal
40	CLK	I	Data lock signal
41 } 44	DATA0 } DATA3	I	Key scan signal
45 } 52	NC	I	Not connected
53	OSC2	I	Clock terminal
54	OSC1	I	Clock input
55	X1	I	Optical servo condition input
56	X0	O	(Not used, open)
57	GND	I	GND terminal
58	DMUTE	O	Muting control
59	MDATA	O	Command data output
60	MCLK	O	Data clock output (command clock)
61	MLD	O	Data output (command load)
62	DOUTON	O	Optical output control signal (Not used, open)
63	EMPH	O	Emphasis signal output
64	NC	—	Not connected

(a) EFM Signal

The RF signal is an analog, continuous signal as shows in figure 17-28. For succeeding digital signal processing, the RF signal is wave-shaped by the data slicing circuit to a digital form signal. The signal output from the data slicing circuit is the EFM signal

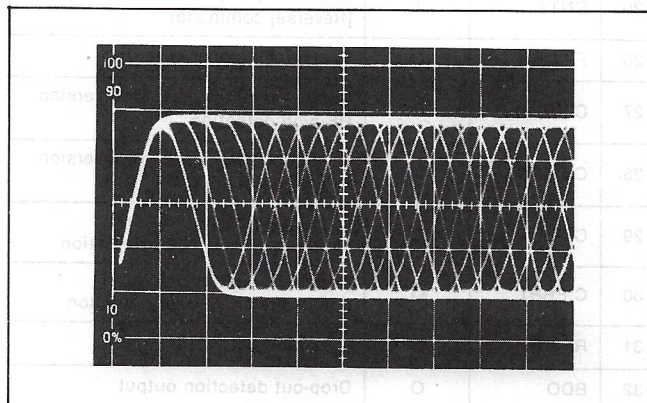


Figure 17-28 RF Signal

(b) PCK Signal

The PCK signal is a clock signal in synchronous with the EFM signal. The EFM signal includes a jitter component to some degree. Therefore the clock signal in synchronous with the EFM signal is required for transmitting the EFM signal exactly to the digital signal processing circuit.

The PLL circuit receives the EFM signal and forms the PCK signal. Figure 17-29 shows the timing chart of the EFM signal and the PCK signal.

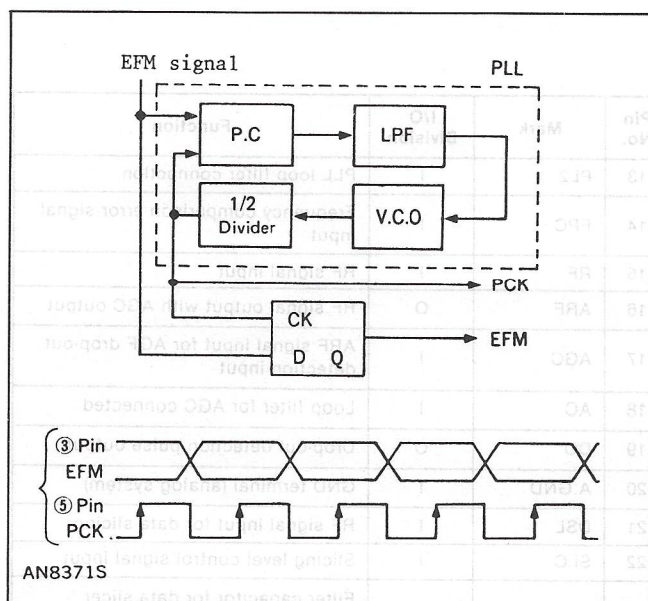


Figure 17-29 EFM and PCK Signal

(c) SRF Signal

The SRF signal is a signal obtained by data-slicing the RF signal. The DO signal is the detection signal of dropout in the RF signal (DO signal is active "High").

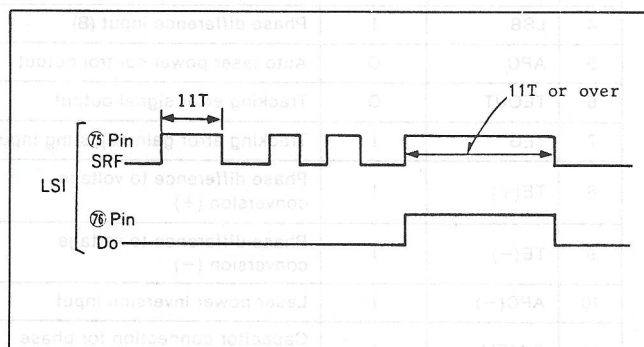


Figure 17-30 SRF Signal

(d) FPC Signal

The FPC signal is the VCO control signal of the PLL circuit and is fed back pin 78 of IC-301.

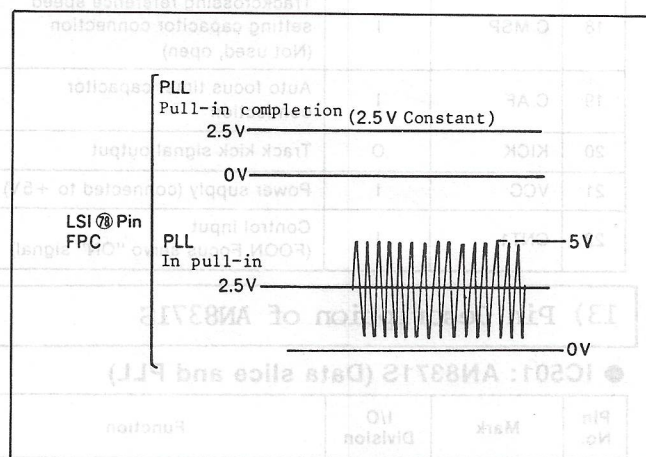


Figure 17-31 FPC Signal

12) Pin Description of AN8370S

● IC101: AN8370S (Optical Servo Control)

Pin No.	Mark	I/O Division	Function
1	VEE	I	Power supply (connected to -5V)
2	LSA	I	Phase difference input (A)
3	GND	I	GND terminal
4	LSB	I	Phase difference input (B)
5	APC	O	Auto laser power control output
6	TEOUT	O	Tracking error signal output
7	TEG	I	Tracking error gain adjusting input
8	TE(+)	I	Phase difference to voltage conversion (+)
9	TE(-)	I	Phase difference to voltage conversion (-)
10	APC(-)	I	Laser power inversion input
11	C.MEM	I	Capacitor connection for phase difference memory
12	APC(+)	I	Laser power non inversion input
13	VREFE	O	Reference current generation
14	SENSE	O	Selector output (track-crossed)
15	HIN	I	Tracking hold circuit input
16	HOUT	O	Tracking hold circuit output
17	SPCNT	O	Trackcrossing speed control output (Not used, open)
18	C.MSP	I	Trackcrossing reference speed setting capacitor connection (Not used, open)
19	C.AF	I	Auto focus timer capacitor connection
20	KICK	O	Track kick signal output
21	VCC	I	Power supply (connected to +5V)
22	CNT1	I	Control input (FOON Focus servo "ON" signal)

Pin No.	Mark	I/O Division	Function
23	CNT2	I	Control input (TRON Tracking servo ON signal)
24	CNT3	I	Control input (KICKF Kick direction [Foward] command)
25	CNT4	I	Control input (KICKR Kick direction [Reverse] command)
26	F.LOCK	O	Focus lock signal output
27	C.FBDO	O	Capacitor connection for inversion RF high detection
28	C.SBDO	O	Capacitor connection for inversion RF low detection
29	C.SBRT	O	Capacitor connection for non-inversion RF slow detection
30	C.FBRT	O	Capacitor connection for non-inversion RF fast detection
31	RF OUT	O	RF signal output
32	BDO	O	Drop-out detection output
33	RF IN	I	RF signal input
34	S.OUT	O	Focus search signal output
35	C.LW	I	Capacitor connection for triangular wave generation
36	FE.OUT	O	Focus error signal output
37	FEG	I	Focus error gain adjusting input
38	FE.REF	I	Focus error comparison voltage generation
39	PDB	I	Photo detection current input (B)
40	IVB	O	Current/voltage conversion (B)
41	IVA	O	Current/voltage conversion (A)
42	PDA	I	Photo detection current input (A)

13) Pin Description of AN8371S

● IC501: AN8371S (Data slice and PLL)

Pin No.	Mark	I/O Division	Function
1	VEE	I	Power supply (connected to -5V)
2	SRF	O	RF signal output data-sliced into digital value
3	EFM	O	EFM signal output synchronized with PCK
4	D.GND	I	GND terminal (digital system)
5	PCK	O	Clock output extracted from SRF
6	VCC	I	Power supply (connected to +5V)
7	VA	I	VCO free run frequency adjusting current input (not connected)
8, 9	VC1, 2	I	Capacitor connection for VCO oscillator frequency
10	VR	I	Resistor connection for VCO oscillator frequency
11	PD	I	Capacitor connection for PLL DO protection
12	PL1	I	PLL loop filter connection

Pin No.	Mark	I/O Division	Function
13	PL2	I	PLL loop filter connection
14	FPC	I	Frequency comparison error signal input
15	RF	I	RF signal input
16	ARF	O	RF signal output with AGC output
17	AGC	I	ARF signal input for AGF drop-out detection input
18	AC	I	Loop filter for AGC connected
19	DO	O	Drop-out detection pulse output
20	A.GND	I	GND terminal (analog system)
21	DSL	I	RF signal input for data slicing
22	SLC	I	Slicing level control signal input
23	FC1	I	Filter capacitor for data slicer connected
24	FC2	I	Filter capacitor for data slicer connected

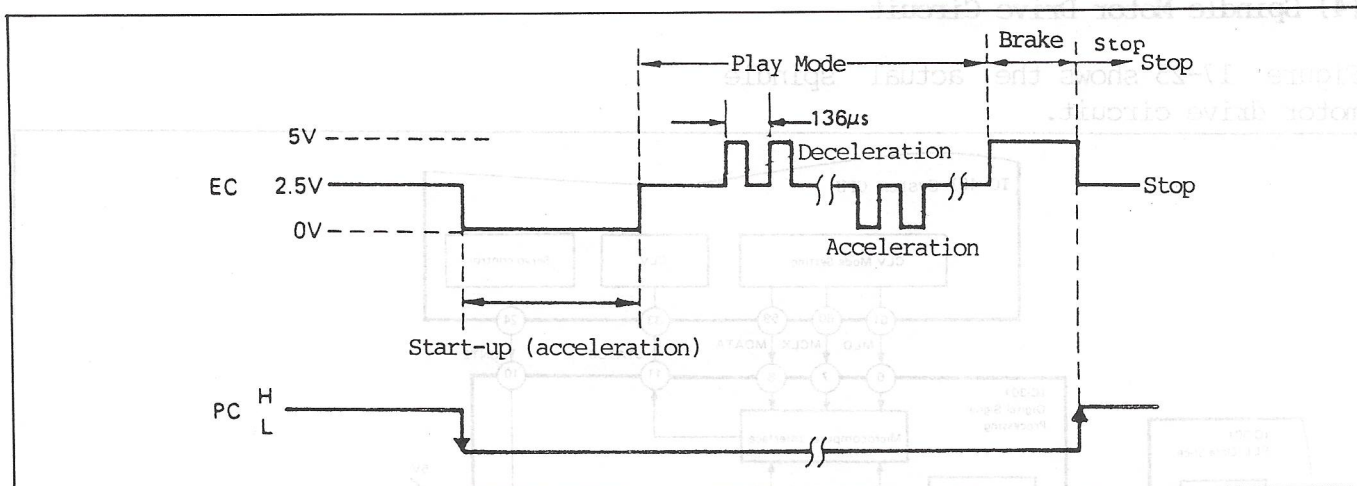


Figure 17-26

When positive pulses are generated, the deceleration mode is selected, and when negative

pulses are generated, the acceleration mode is selected.

11) PLL Servo Circuit

A clock component is extracted from the RF signal which is read off the disc. The clock is used to regenerate the digital musical signal. (The CLV servo controls the spindle motor so that the digital signal read off the disc is synchronized with a reference quartz-controlled clock. It includes a coarse servo which starts up until the PLL is locked, and a fine servo which brings the read signal into phase

synchronization with the reference clock)

The PLL and CLV servo are in a parent-child relationship. The CLV servo absorbs slow speed variation resulting from the disc being off-center or other factors, whereas the PLL absorbs finer jitter, so that an uninterrupted digital signal is regenerated from the disc pits.

Figure 17-27 shows block diagram of PLL circuit.

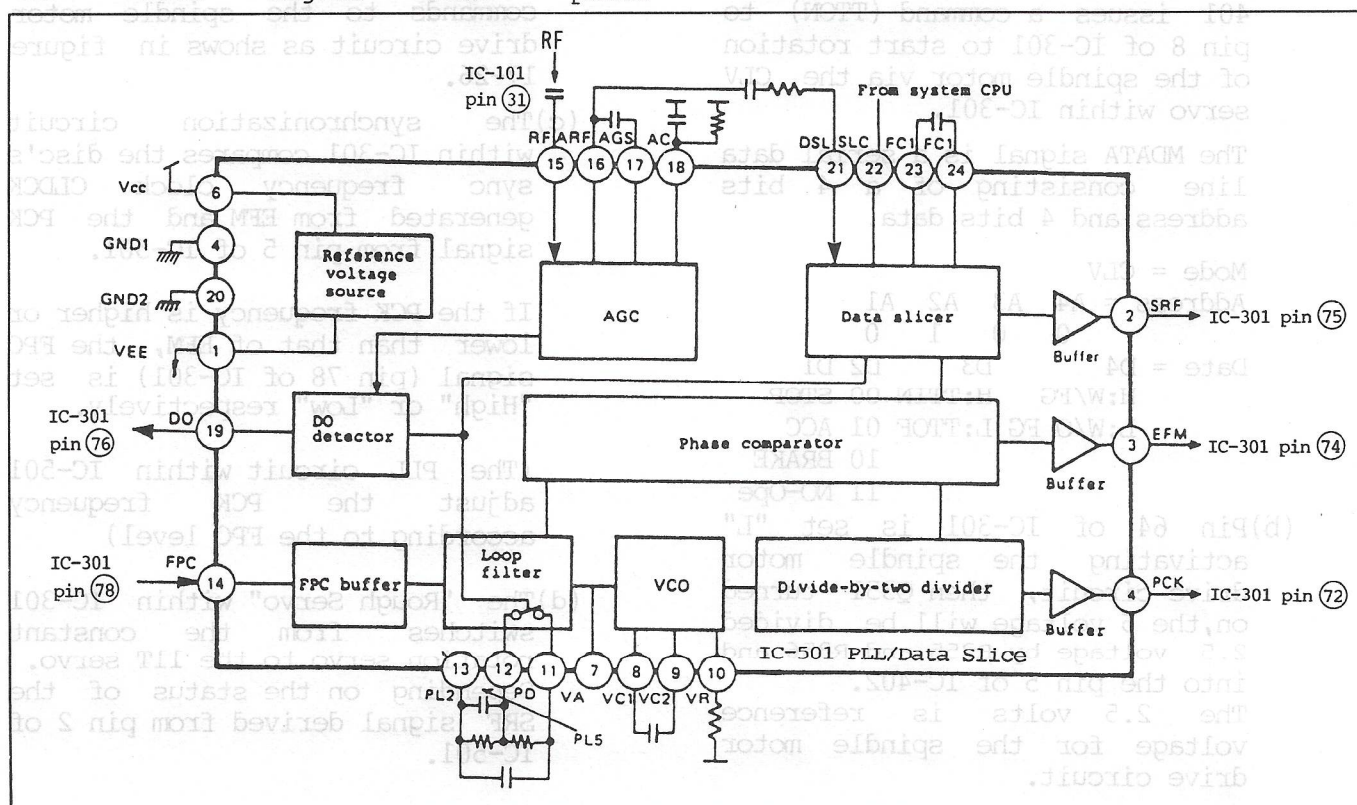


Figure 17-27 Block Diagram of AN8371S

(4) Spindle Motor Drive Circuit

Figure 17-25 shows the actual spindle motor drive circuit.

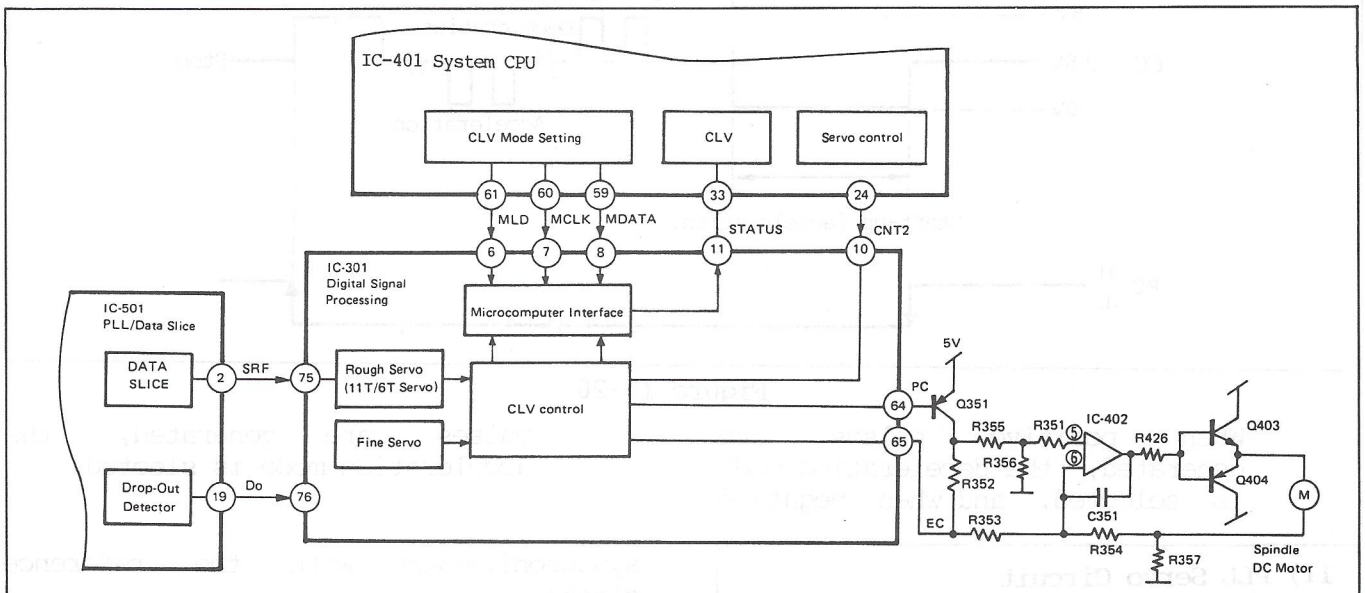


Figure 17-25 Spindle Motor Drive Circuit

During the process, from start-up to CLV, the servo steps through three stages:

- Constant Rotation (Start-up)
- 11T Servo (Coarse Control)
- CLV Servo (Fine Control)

1). Operation from Start-Up to Play:

- The MDATA line from pin 59 of IC-401 issues a command (TTON) to pin 8 of IC-301 to start rotation of the spindle motor via the CLV servo within IC-301.

The MDATA signal is a serial data line consisting of a 4 bits address and 4 bits data.

Mode = CLV

Address = A4 A3 A2 A1
0 0 1 0

Date = D4 D3 D2 D1
H:W/FG H:TTIN 00 STOP
L:W/O FG L:TTOF 01 ACC
10 BRAKE
11 NO-Ope

- Pin 64 of IC-301 is set "L" activating the spindle motor drive circuit, then Q351 turned on, the 5 voltage will be divided 2.5 voltage by R355 and R356 and into the pin 5 of IC-402. The 2.5 volts is reference voltage for the spindle motor drive circuit.

The 11T servo mode is selected when the tracking servo is active (TRON signal/CONT2 is "Low") and the PLL is unlocked.

- Until the frequencies are synchronized, the EC signal at pin 65 of IC-301 provides acceleration or deceleration commands to the spindle motor drive circuit as shows in figure 17-26.

- The synchronization circuit within IC-301 compares the disc's sync frequency clock CLDCK generated from EFM and the PCK signal from pin 5 of IC-501.

If the PCK frequency is higher or lower than that of EFM, the FPC signal (pin 78 of IC-301) is set "High" or "Low" respectively.

(The PLL circuit within IC-501 adjust the PCK frequency according to the FPC level)

- The "Rough Servo" within IC-301 switches from the constant rotation servo to the 11T servo. Depending on the status of the SRF signal derived from pin 2 of IC-501.

(1) Principle of the Linear Motor

As can be seen from figure 17-20, a linear DC motor is in fact a coil suspended in a magnetic field.

When a current I is sent through the coil, a force will be applied onto the coil and the latter will move accordingly. The direction of the motion according to the force applied, is determined by Fleming's left hand rule.

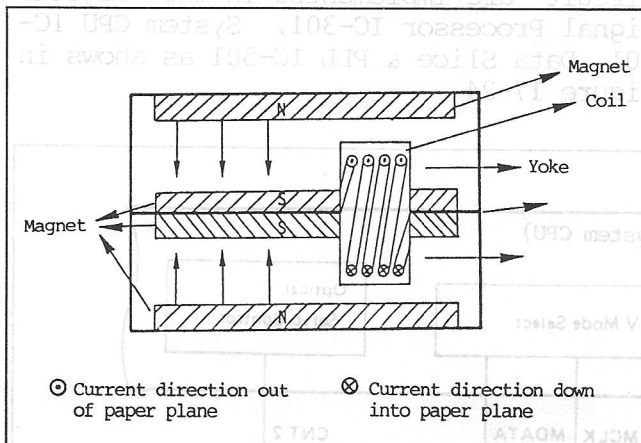


Figure 17-22 Linear Motor Principle

The index finger points in the direction of the magnetic field, whereas the middle finger points in the direction of the current through the coil, and the thumb indicates the direction of the motion of the coil as shown in figure 17-22 and figure 17-23.

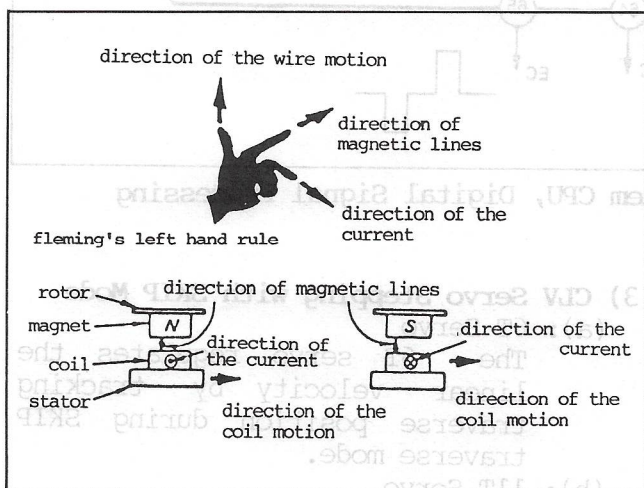


Figure 17-23

10) Spindle (CLV) Servo Circuit

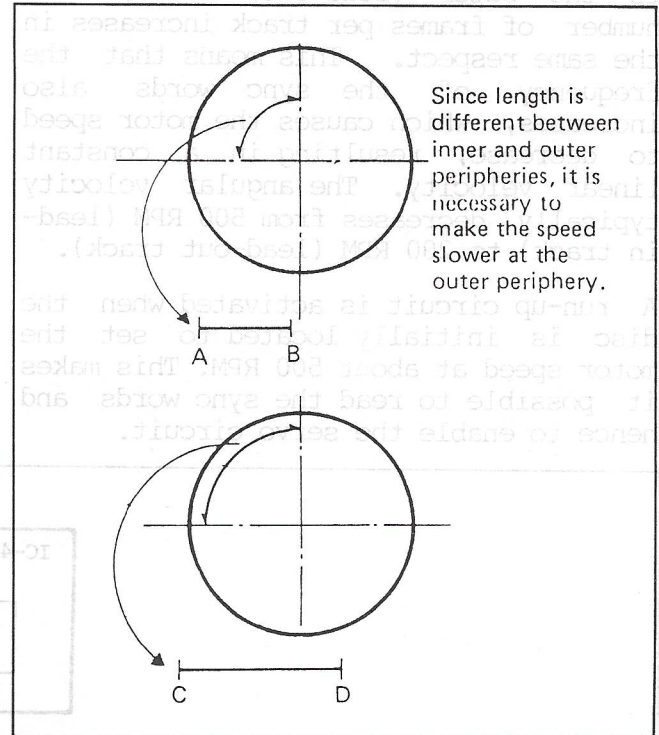


Chart 17-1 Arc Length at Different Radii

What is a Linear Velocity?

The compact disc uses a constant linear velocity (CLV) drive system. This means that the disc speed changes to maintain a constant reading speed approx 1.3 m/s. In figure 17-1, shows the distance from A to B is shorter than from C to D.

To read this data at a constant speed the disc must rotate slower at the outer periphery. The range is 500 rpm through 200 rpm from inner to outer periphery. The proper speed is maintained by reading the synchronizing signal (7.35KHz) and comparing it to a reference signal.

As pointed out before, the data on the disc is recorded in blocks, which we called "frame". Each frame starts with a unique sync word, which is used to control the spindle motor.

The frequency of these sync words is compared against a fixed frequency (derived from a crystal oscillator) in a phase comparator and the motor is driven according to any frequency or phase difference.

As the length of the tracks increases linearly from the linner (lead-in) track to the outer (lead-out) track, the number of frames per track increases in the same respect. This means that the frequency of the sync words also increases, which causes the motor speed to decrease, resulting in a constant linear velocity. The angular velocity typically decreases from 500 RPM (lead-in track) to 200 RPM (lead-out track).

A run-up circuit is activated when the disc is initially located to set the motor speed at about 500 RPM. This makes it possible to read the sync words and hence to enable the servo circuit.

(1) Correlation with IC-301, IC-401, IC-501

The CLV servo circuit controls disc rotation to a constant linear velocity. It includes three servo modes, i.e. CVL (fine), the 11T, and 6T.

** 6T = Traverse Tracking

** 11T = Coarse Servo Control

** CLV = Provides Fine Servo Control

The major block of the Spindle Servo Circuit are implemented in the Digital Signal Processor IC-301, System CPU IC-401, Data Slice & PLL IC-501 as shows in figure 17-24

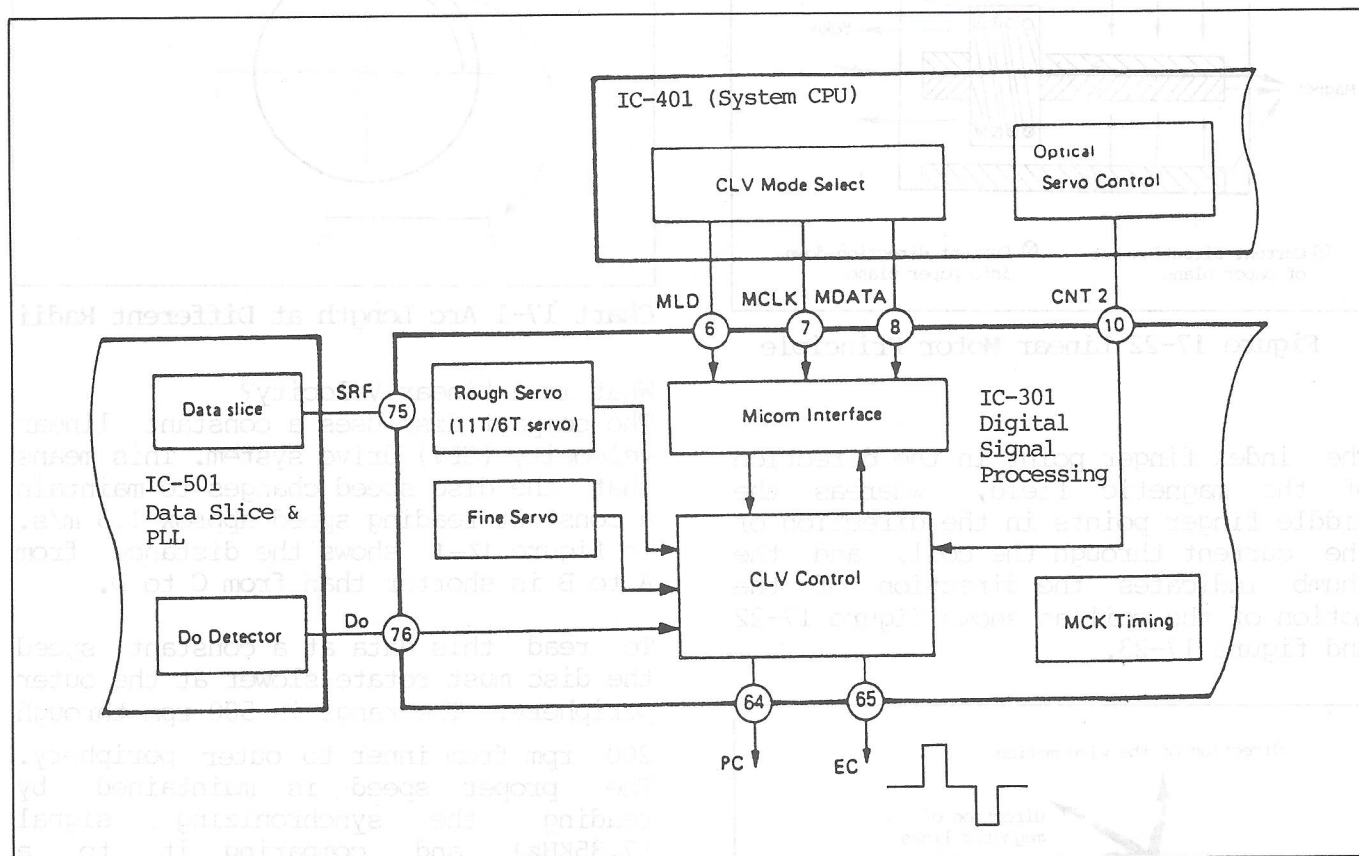


Figure 17-24 Connection Diagram of System CPU, Digital Signal Processing and PLL/Data Slice

(2) CLV Servo Stepping with Play Mode

(a): 11T Servo

The 11T servo controls the SRF signal (data-sliced RF signal) so that the signal's maximum period is 11T time the reference clock (1T=4.3MHz) period.

(b): CLV Servo

The CLV servo initiates its operation after IC-301 is made ready for internal digital processing by the 11T servo.

(3) CLV Servo Stepping with SKIP Mode

(a): 6T Servo

The 6T servo regulates the linear velocity by tracking traverse position during SKIP traverse mode.

(b): 11T Servo

Same as above (a)

(c): CLV Servo

Same as above (b)

The linear motor consists of a coil wound around a yoke plate and a magnet at both end. The coil undergoes an infkluence from a force filed and will

move across the yoke depending on the direction of the current through the coil.

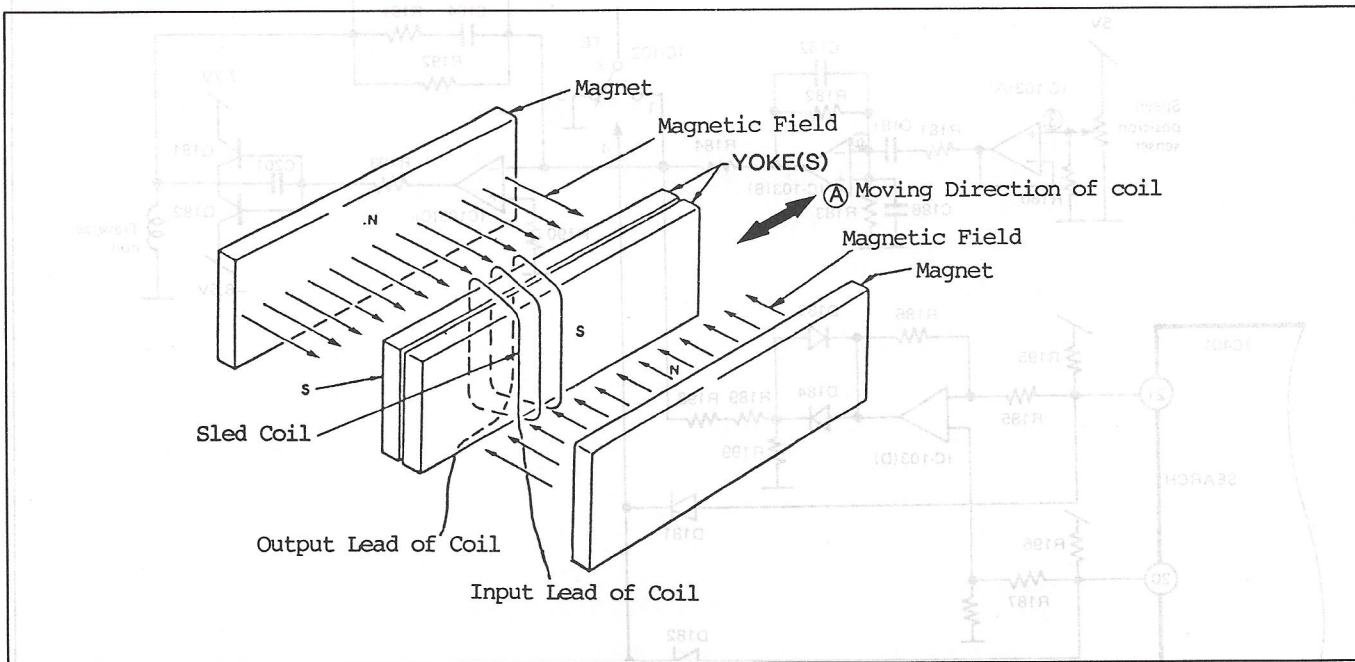


Figure 17-20 Liner Motor Principle

The traverse servo circuit receives the tracking error signal at pin 2 of IC-102, The servo circuit will output a traverse signal to correct the position of the optical pickup to keep it on the pit track.

To perform a 1, 10, or 100 track jump, (SKIP) the command from the system IC-401 will be controlled at IC-103, the servo control, and this circuit will drive the traverse motor. This allows a 1, 10, or 100 track jump to be executed very fast and smooth.

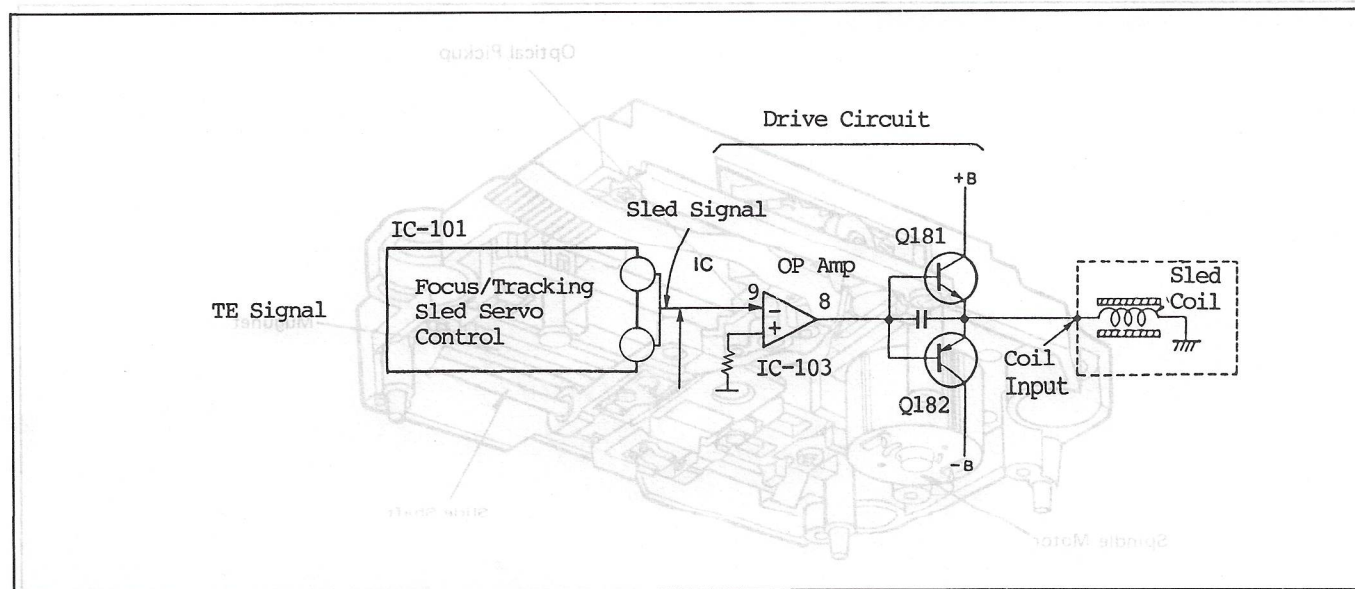


Figure 17-21 Linear Motor Servo Circuit Principle

The figure 17-18 shows traverse servo circuit.

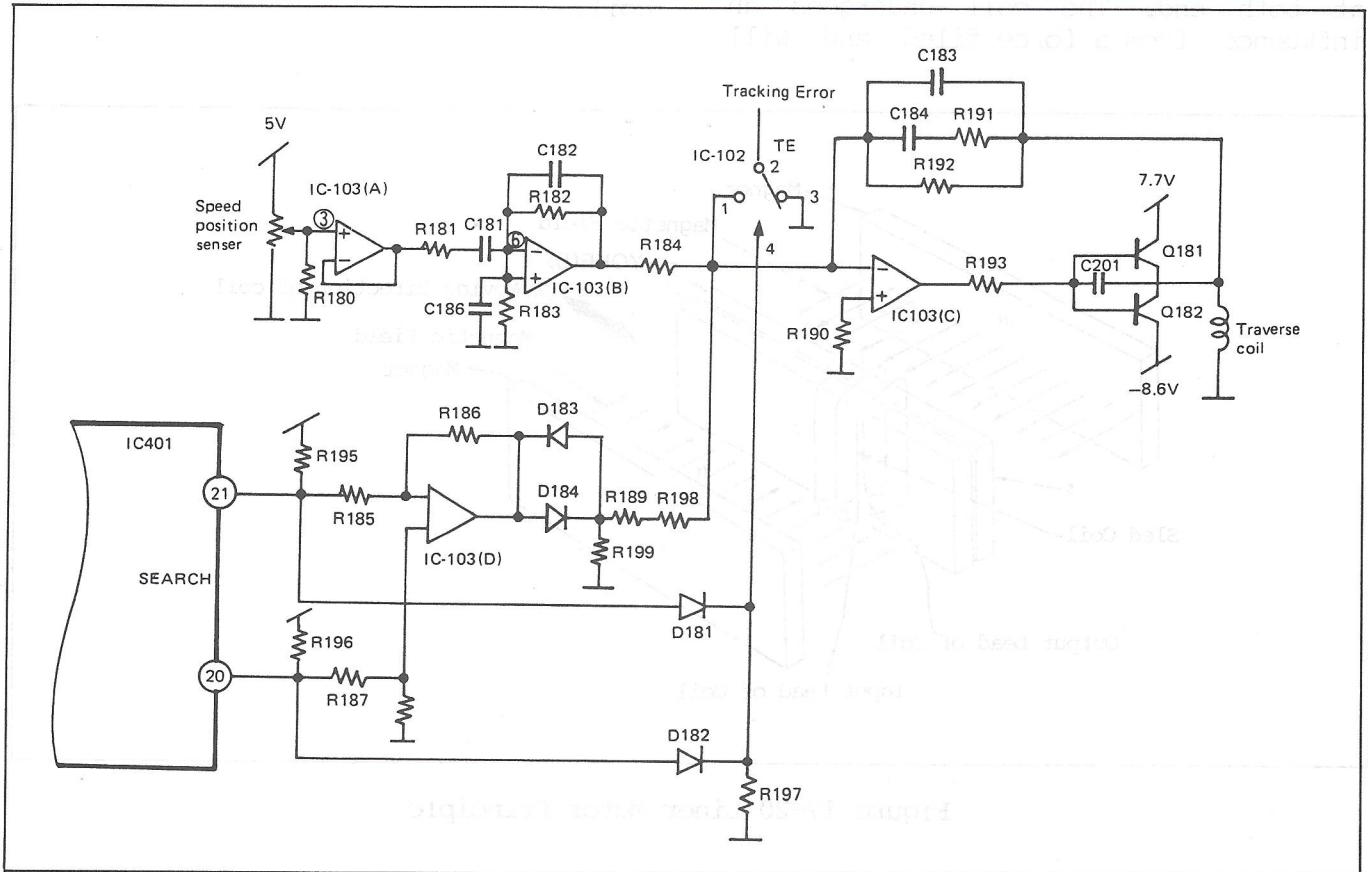


Figure 17-18 Traverse Servo Circuit

9) Linear Motor Operation

A linear motor is used for the optical pickup transportation mechanism in the new CD players. The figure 17-19 shows

the optical pickup transportation mechanism. The linear motor drive coil (traverse coil) is a part of the optical pickup and performs transportation operation.

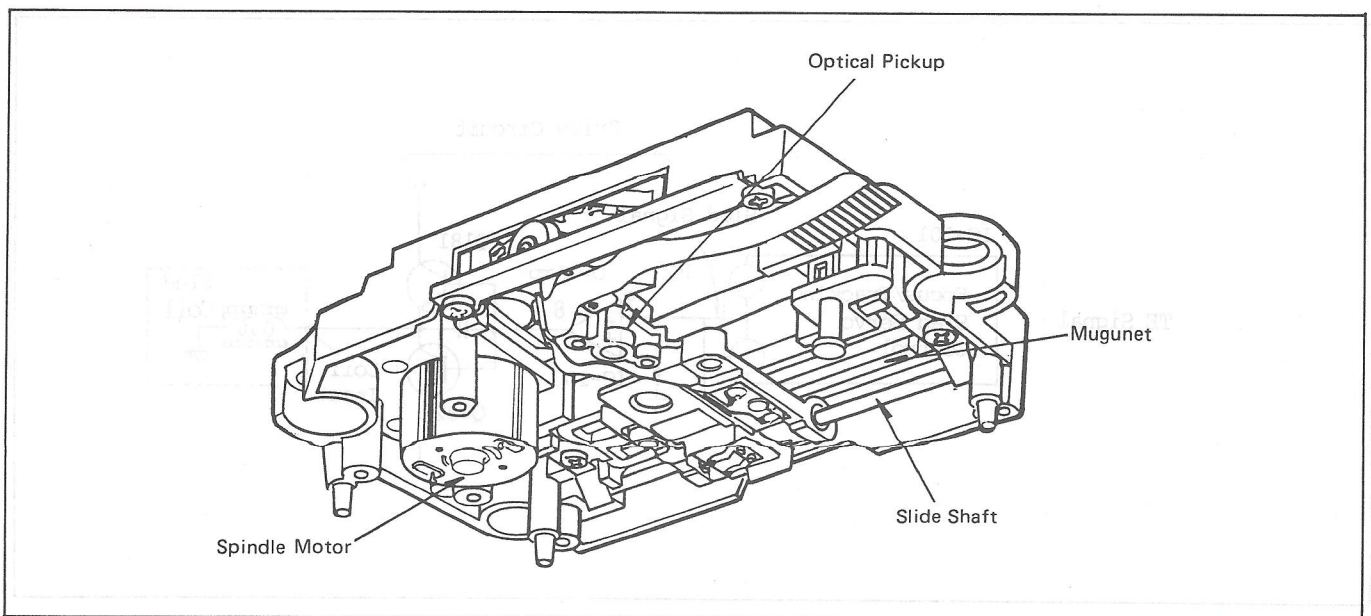


Figure 17-19 Optical Pickup Transportation Mechanism

The phase of the CROSS signal leads in the KICK FWD mode and lags in the KICK REV mode. The brake direction is determined by the detection of the above phase difference.

(4) Tracking Drive Circuit

The TE signal appearing at pin 16 of IC-101 is applied to an amplifier (with low-frequency complicity) comprised of IC-106, R105, C164, and R167. The output of the amplifier couples to the current feedback type tracking drive circuit

comprised of Q161, and Q162 which then drive the tracking drive coil.

7) Relation with Servo IC and System CPU

The figure 17-16 shows the interconnection diagram of servo IC-101 (AN8370S) and system IC-401. There are four buses for carrying signals from the system IC to AN8370S and two buses for carrying signal from AN8370S to the system IC.

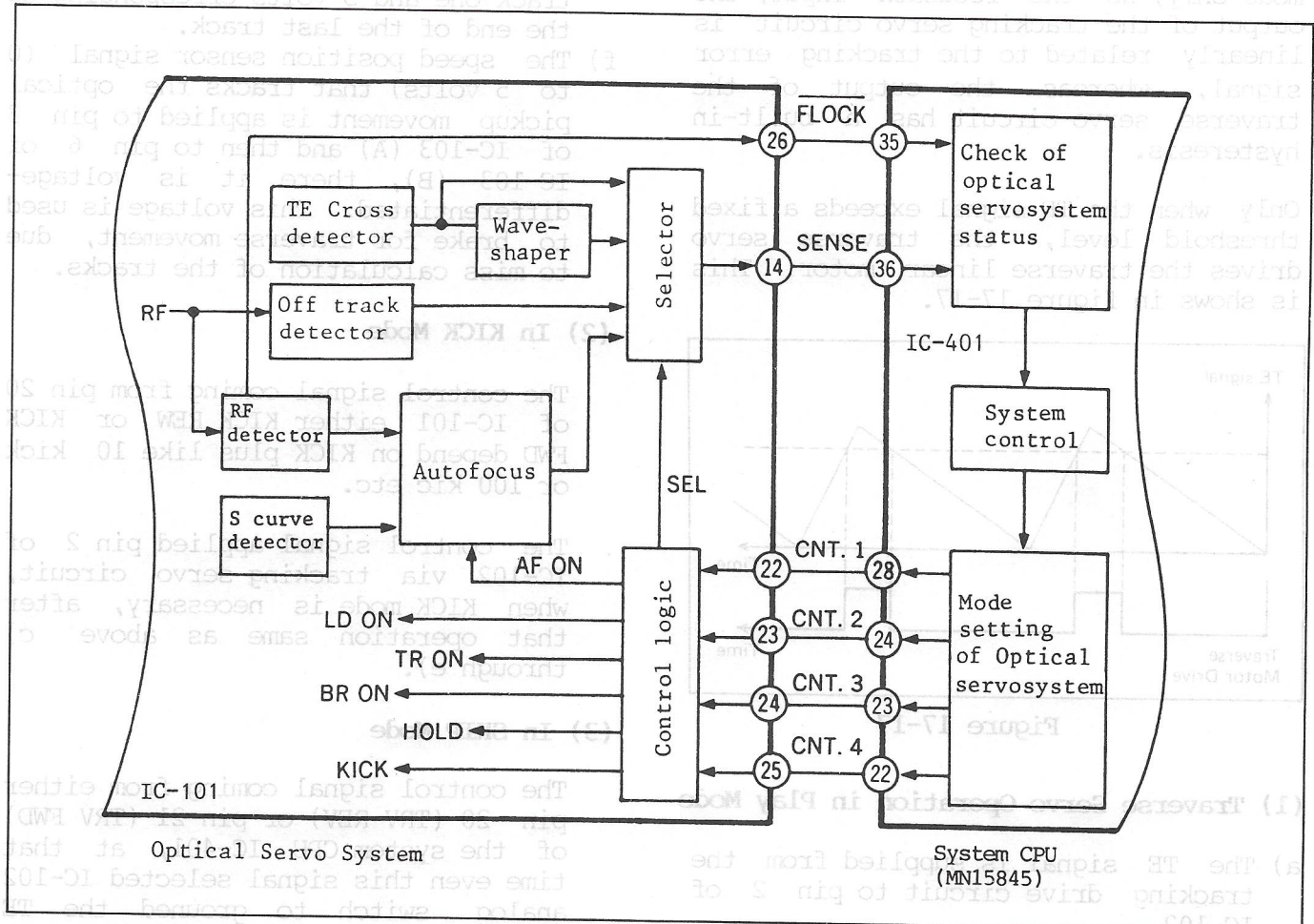


Figure 17-16 Connection Diagram of System CPU and Optical Servo System

- (a) Output signal from system:
- Control logic signals (CNT1 -CNT4)

- (b) Input signal from system:
- RF Detection signal (FLOCK).....
 - ...Active "L"

- Servo system status signal (SENSE)..
- Type of SENCE signal:
- (1)Off-track signal
- (2)Tracking cross signal
- (3)Waveshaped tracking cross signal
- (4)Focus S curve detection signal.

8) Traverse Servo Circuit

The 2-axis device can move horizontally and thus follows a limited number of tracks (fine tracking), Another servo circuit, which called the traverse servo circuit, is used to move the complete optical unit across the disc surface (course tracking).

This servo circuit uses the same tracking error signal due to normal playing mode (insted of the SKIP mode only) as the feedback input, the output of the tracking servo circuit is linearly related to the tracking error signal, whereas the output of the traverse servo circuit has a built-in hysteresis.

Only when the TE signal exceeds a fixed threshold level, the traverse servo drives the traverse linear motor. This is shows in figure 17-17.

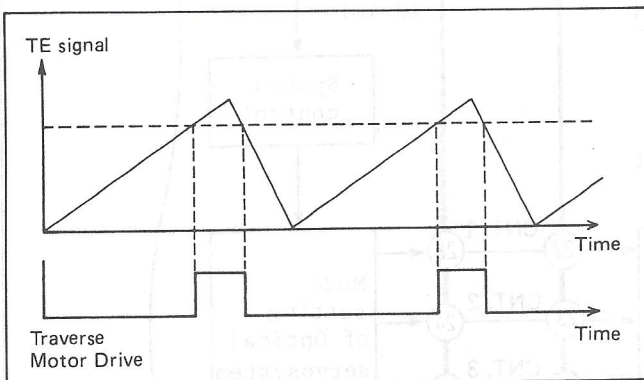


Figure 17-17

(1) Traverse Servo Operation in Play Mode

- The TE signal is supplied from the tracking drive circuit to pin 2 of IC-102.
- The TE signal selected by IC-102 due to play mode, incase of the search mode, the TE signal is ground by IC-102 pin 2 through pin 3.
- The incase of due to play mode, the TE signal into the IC-103 (C) pin 9 via IC-102 pin 1.
- The output of IC-103 (C) pin 8 is used to set the base potential of the traverse driving transistors Q181 and Q182 to determine the direction of the traverse movement.

- The base potential of Q181 and Q182 is set to positive or negative, according to the TE signal polarity to control the traverse drive current, so that in turn tracking error is minimized.
- The traverse drive mechanism has a speed position sensor (potentiometer) which tracks the optical pickup movement to provide pickup position information. The voltage range which varies with the speed position sensor, is set between 0 volt corresponding to the beginning of track one and 5 volts crresponding to the end of the last track.
- The speed position sensor signal (0 to 5 volts) that tracks the optical pickup movement is applied to pin 3 of IC-103 (A) and then to pin 6 of IC-103 (B), there it is voltage-differentiated. This voltage is used to brake for traverse movement, due to miss calculation of the tracks.

(2) In KICK Mode

The control signal coming from pin 20 of IC-101 either KICK REW or KICK FWD depend on KICK plus like 10 kick or 100 kic etc.

The control signal applied pin 2 of IC-102 via tracking servo circuit, when KICK mode is necessary, after that operation same as above c) through e).

(3) In SKIP Mode

The control signal coming from either pin 20 (TRV REV) or pin 21 (TRV FWD) of the system CPU IC-401, at that time even this signal selected IC-102 analog switch to ground the TE signal in normal play mode and, instead, applies either the FWD or REV SKIP signal to IC-103 (D) to drive the traverse mechanism. The subsequent operations of the traverse mechanism are idetical to above step c) through e) in the previous paragraphs.

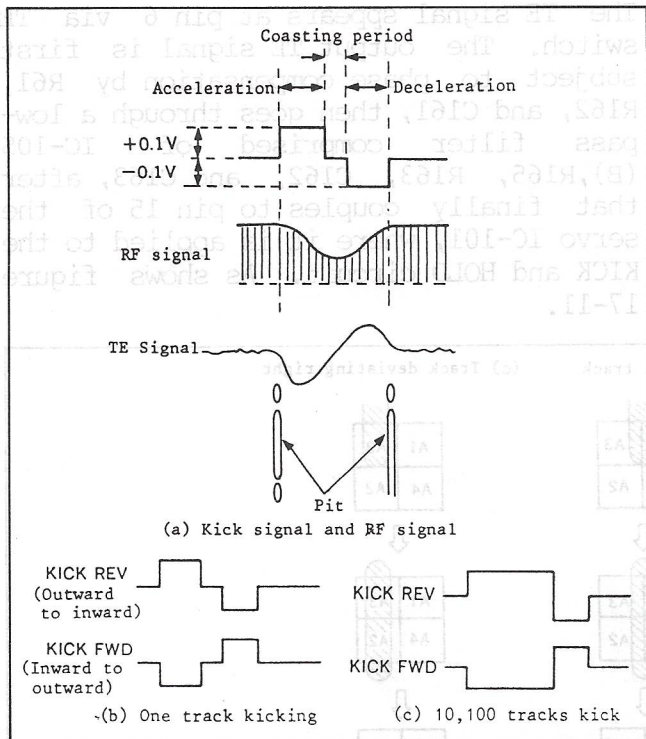


Figure 17-12 KICK Signal and RF Signal

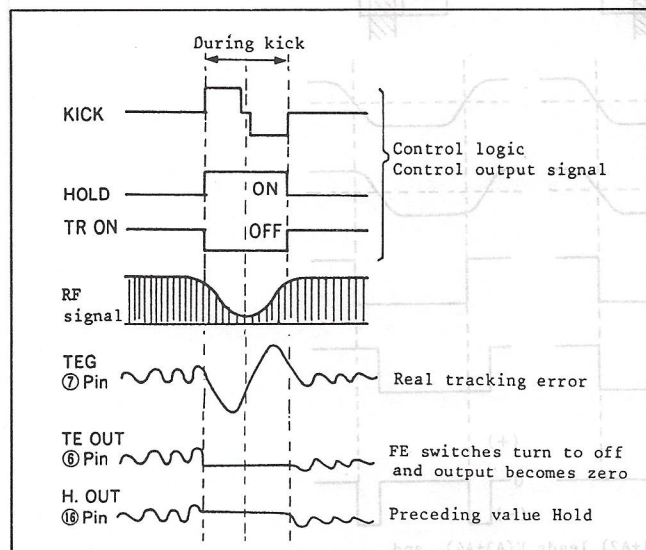


Figure 17-13 PAUSE KICK Mode

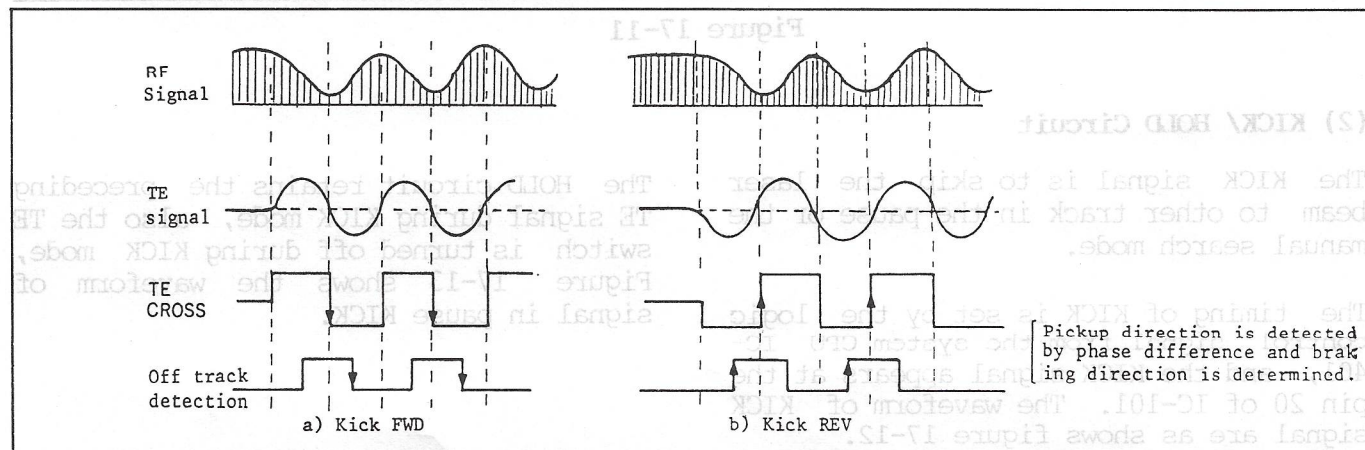


Figure 17-15 KICK Direction Discrimination

(3) Braking Circuit

The brake circuit brakes the tracking actuator after quick traverse or KICK (10 or 100 track) for rapidly locking the tracking servo. The principle of the brack circuit as shows in figure 17-14.

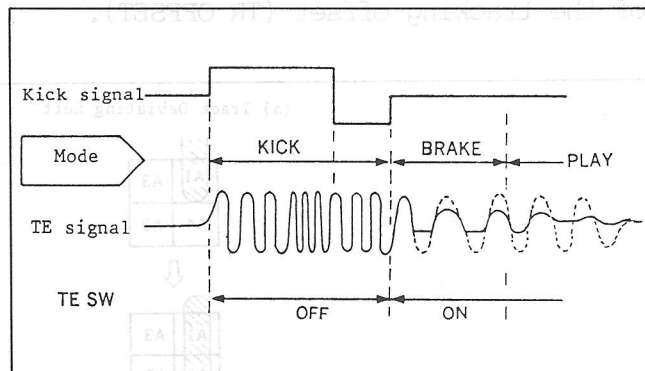


Figure 17-14 Brake Timing

In braking, the TE signal is limited on one side (the continuous line shows in figure 17-14) to absord energy (of the actuator) after KICK.

The figure 17-14 shows braking in the KICK REV mode. In the KICK FWD mode, the braking is made by reversing the polarity of the sideband limiter. Whereas, the brake circuit requires a signal for discriminating the direction of KICK.

The KICK direction discriminating signal is formed by the tracking cross (CROSS) signal and off-track signal as shows in figure 17-15.

The analog tracking error (TE) signal is obtained by smoothing the above mentioned generated pulses with LPF. The above processes are carried out by the phase comparator and LPF.

VR102 is the adjuster of the tracking gain (TR GAIN) and VR103 is the adjuster of the tracking offset (TR OFFSET).

The TE signal appears at pin 6 via TE switch. The output TE signal is first subject to phase compensation by R61, R162, and C161, then goes through a low-pass filter comprised of IC-105 (B), R165, R163, C162, and C163, after that finally couples to pin 15 of the servo IC-101, where it is applied to the KICK and HOLD circuit. As shows figure 17-11.

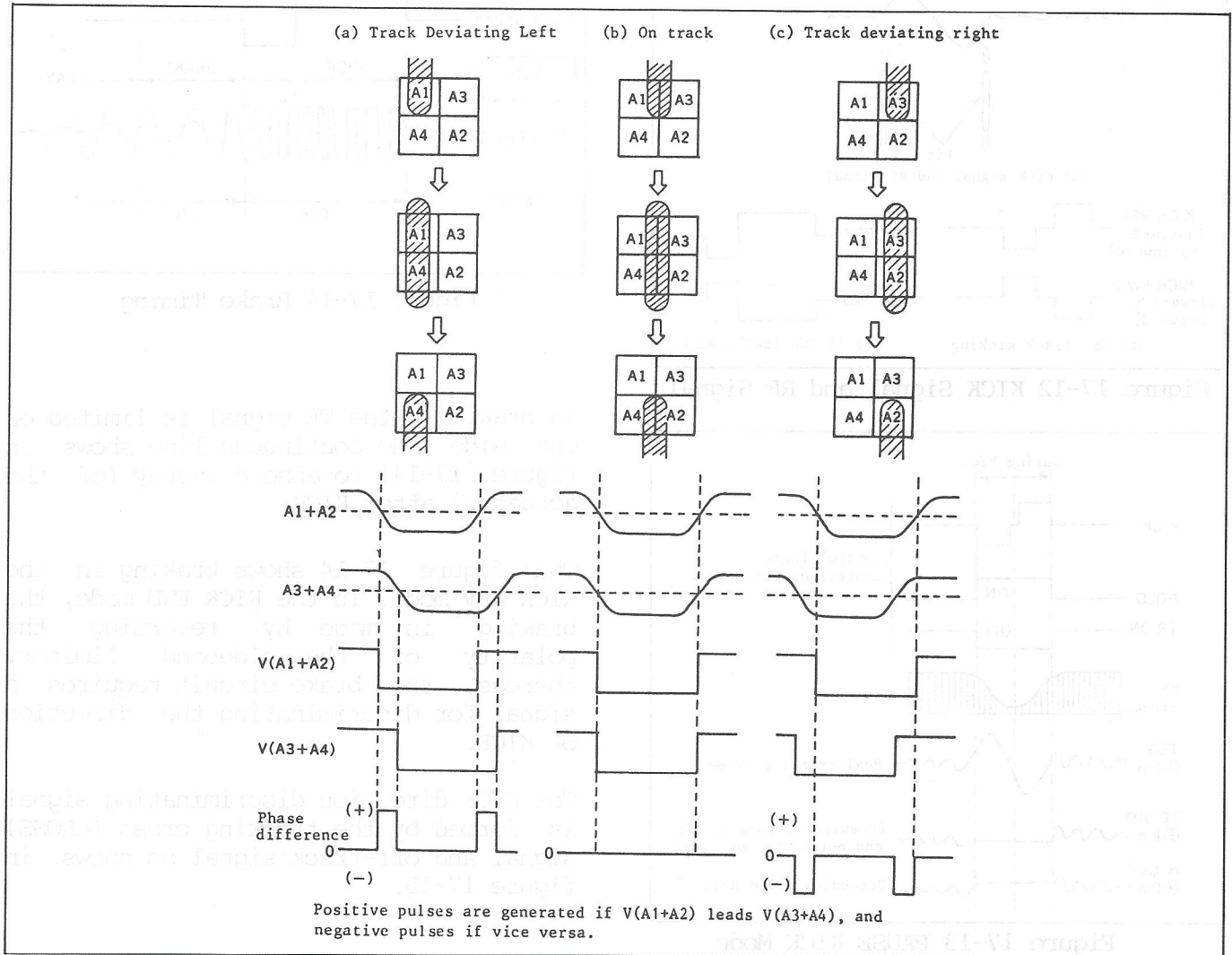


Figure 17-11

(2) KICK/ HOLD Circuit

The KICK signal is to skip the laser beam to other track in the pause or the manual search mode.

The timing of KICK is set by the logic control signal from the system CPU IC-401, and the KICK signal appears at the pin 20 of IC-101. The waveform of KICK signal are as shows figure 17-12.

The HOLD circuit retains the preceding TE signal during KICK mode, also the TE switch is turned off during KICK mode, Figure 17-13 shows the waveform of signal in pause KICK.

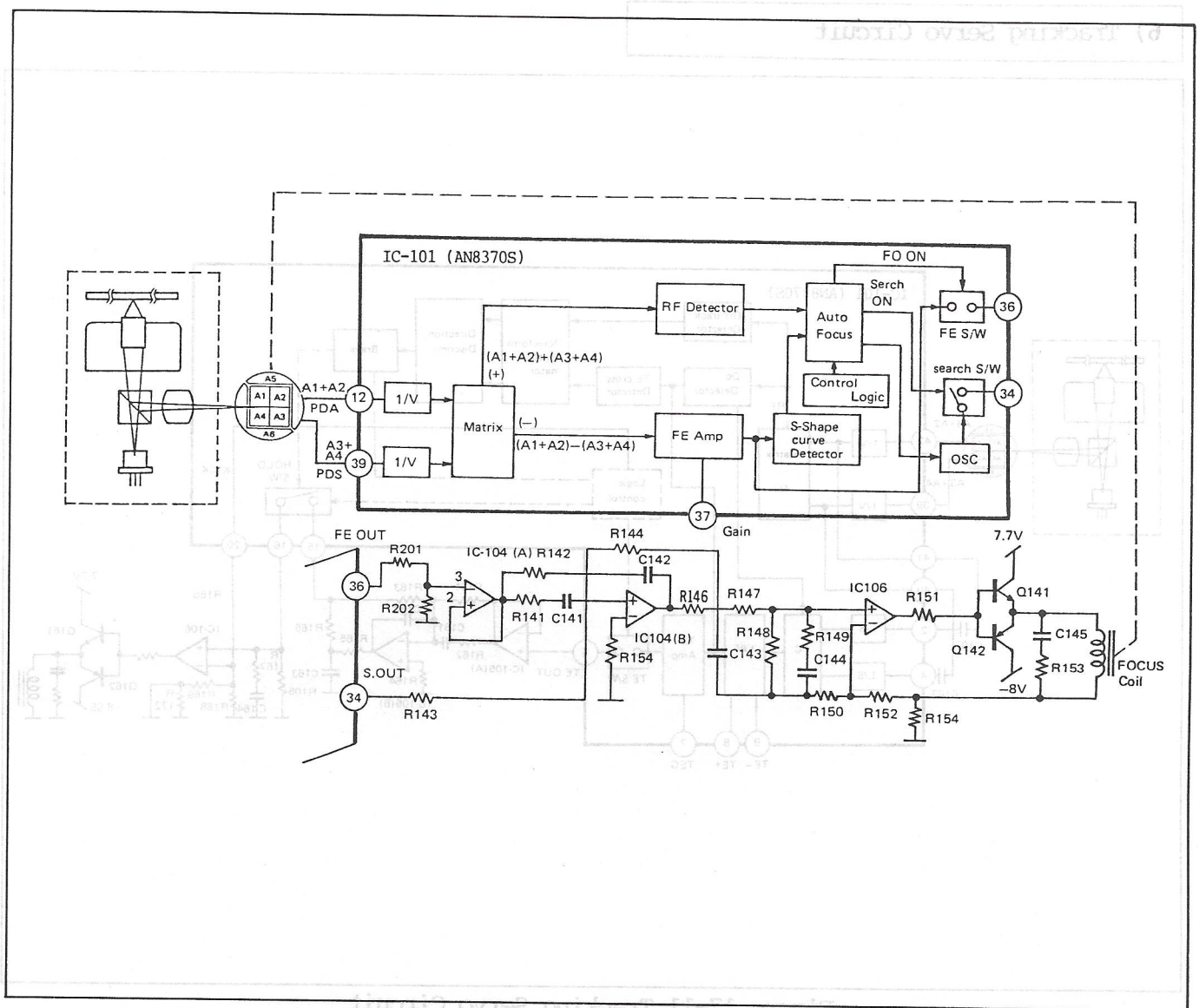


Figure 17-10 Focus Servo Circuit

The tracking servo circuit accepts parallel 4 bits control data transferred from the system CPU of IC-401 and drives the entire tracking servo system, according to the CPU commands, after focus servo is established.

(1) Tracking Error Circuit

The tracking error circuit applies the tracking error signal as in the preceding paragraph to the pin 2 and pin 4 of IC-101 via C102 and C103, and shapes the waveforms of the signals by the level shifter.

As outlined in the para. difference in phase between the two signals is detected and pulse in proportion to the phase difference are generated as the tracking by the polarity of the pulses.

After focus servo is established, the tracking error (TE) signal is created by applying the diagonal PD output signals (A1+A2), (A3+A4), into the servo system IC-101 pin 39 and pin 42 (same as focus servo).

With this, the tracking servo circuit identifies whether the laser beam is on track or off track, by picking up an error signal that corresponds to the off-center span of the disc. The internal control logic closes the tracking servo loop switch (TE Switch) when the beam is on track.

6) Tracking Servo Circuit

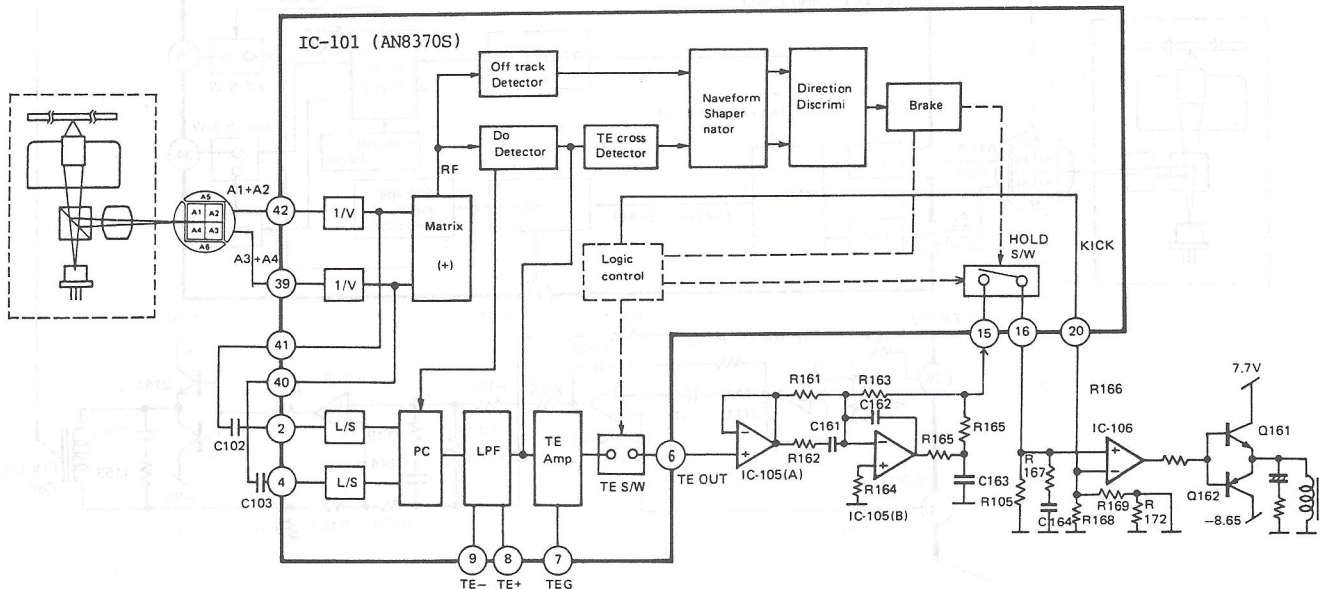


Figure 17-11 Tracking Servo Circuit

The tracking servo circuit accepts parallel 4 bits control data transferred from the system CPU of IC-401 and drives the entire tracking servo system, according to the CPU commands, after focus servo is established.

After focus servo is established, the tracking error (TE) signal is created by applying the diagonal PD output signals (A1+A2), (A3+A4), into the servo system IC-101 pin 39 and pin 42 (same as focus servo).

With this, the tracking servo circuit identifies whether the laser beam is on track or off track, by picking up an error signal that corresponds to the off-center span of the disc. The internal control logic closes the tracking servo loop switch (TE Switch) when the beam is on track.

The specified track is sought with the track signal. Since the TE signal is not needed during the track KICK sequence, the circuit closes the Hold switch to temporarily hold the previous tracking error signal.

(1) Tracking Error Circuit

The tracking error circuit applies the I/V converted signal as in the preceding paragraph to the pin 2 and pin4 of IC-101 via C102 and C103, and shapes the waveforms of the signals by the level shifter.

As outlined in the para. , the difference in phase between the two signals is detected and pulses in proportion to the phase difference are generated as the tracking by the polarity of the pulses.

(1) Autofocus Circuit

The autofocus circuit is operated by the AF ON (autofocus command) signal output from the control logic circuit. Two signals (RF detection signal and S-curve detection signal) are input to the autofocus circuit which regulates the search switch and FE (focus error) switch.

1 At detection of the disc

On the receipt of the AF ON (autofocus on) signal, at first the search switch is turned on and next the focus error switch is turned off. In this situation, the search signal (sawtooth waves) drives the focus coil forcedly to move the lens up and down.

When the distance between the disc and the lens nears the focus length (lock-in focus range), the FE signal shows an "S" curve, with the search switch is turned off and the focus error switch is turned on to close the focus servo loop, as shows in figure 17-9.

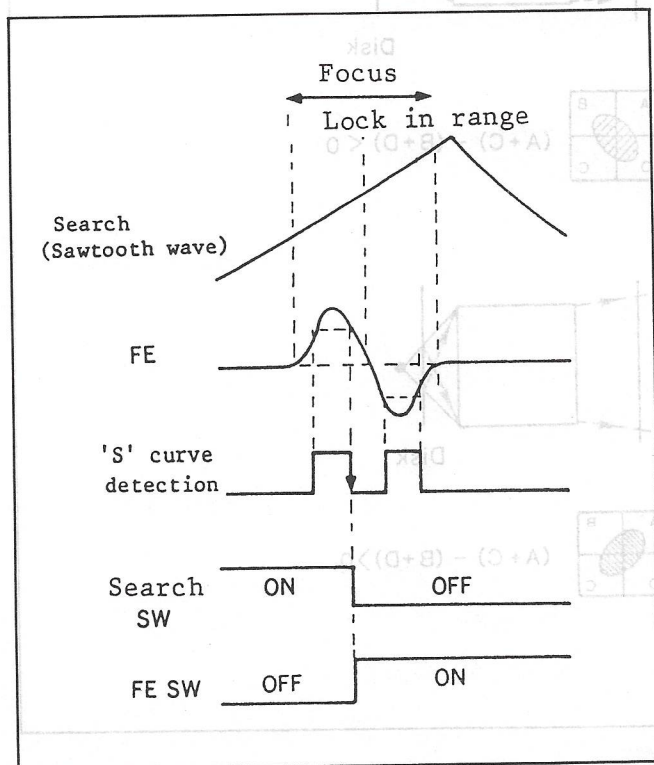


Figure 17-9 Focus On Timing

2 Re-locking in focus

The autofocus circuit checks the servo status by RF detection when the focus servo loop is closed. If the disc deviates from the focal point and RF search oscillation is started. The search oscillation signal is input to the system CPU. Whether or not autofocus is working is known by the voltage at the pin 26 (FLOCK) of IC-101. The voltage level is active "L" when locked in focus.

(2) Focus Error Circuit

The sums of output signals from pair of diagonal segments each (A1+A2) and (A3+A4) of the 4 segment PD are applied to the pin 39 and 42 of IC-101. The I/V converter converts the output currents of PD to voltages.

The gain is set by an exterior resistor, and VR101 is the best eye adjuster for setting off the sensitivity difference of PD.

The difference between the converted two signal (A1+A2) - (A3+A4) is focus error signal (FE). The FE signal applied to the pin 36 (FE OUT) of IC-101 via the FE amplifier and FE switch, and applied to the pin 3 of IC-104 (A).

VR104 is the adjuster of focus gain (FO GAIN) and VR105 is the adjuster of focus off set (FO OFFSET).

The FE signal applied to the focus drive circuit from pin 36 of IC-101, The first subjected to phase compensation by R142, R141, and C141. It then goes through a low-pass filter comprised of IC-104 (B), R144, C142, and C143, and then an amplifier (with low frequency compensation) comprised of IC-106, R149, C144, and R148.

The output of the amplifier is applied to current-feedback focus drive, Q141 and Q142, which then drives the focusing lens drive coil as required, as shows in figure 17-10.

This servo circuit controls the vertical movement of the 2-axis device and guarantees that the focus point of the laser beam is precisely on the mirror surface of the compact disc.

Figure 17-8 shows the three possible situations and the resultant output from

the main spot detector. This photodetector actually is an array of four photo-diodes, which are A1, A2, A3, and A4. The reflected laser beam hits these four diodes and generates a four error signal:

$$\text{Focus Error (F.E.)} = (A1+A2)-(A3+A4)$$

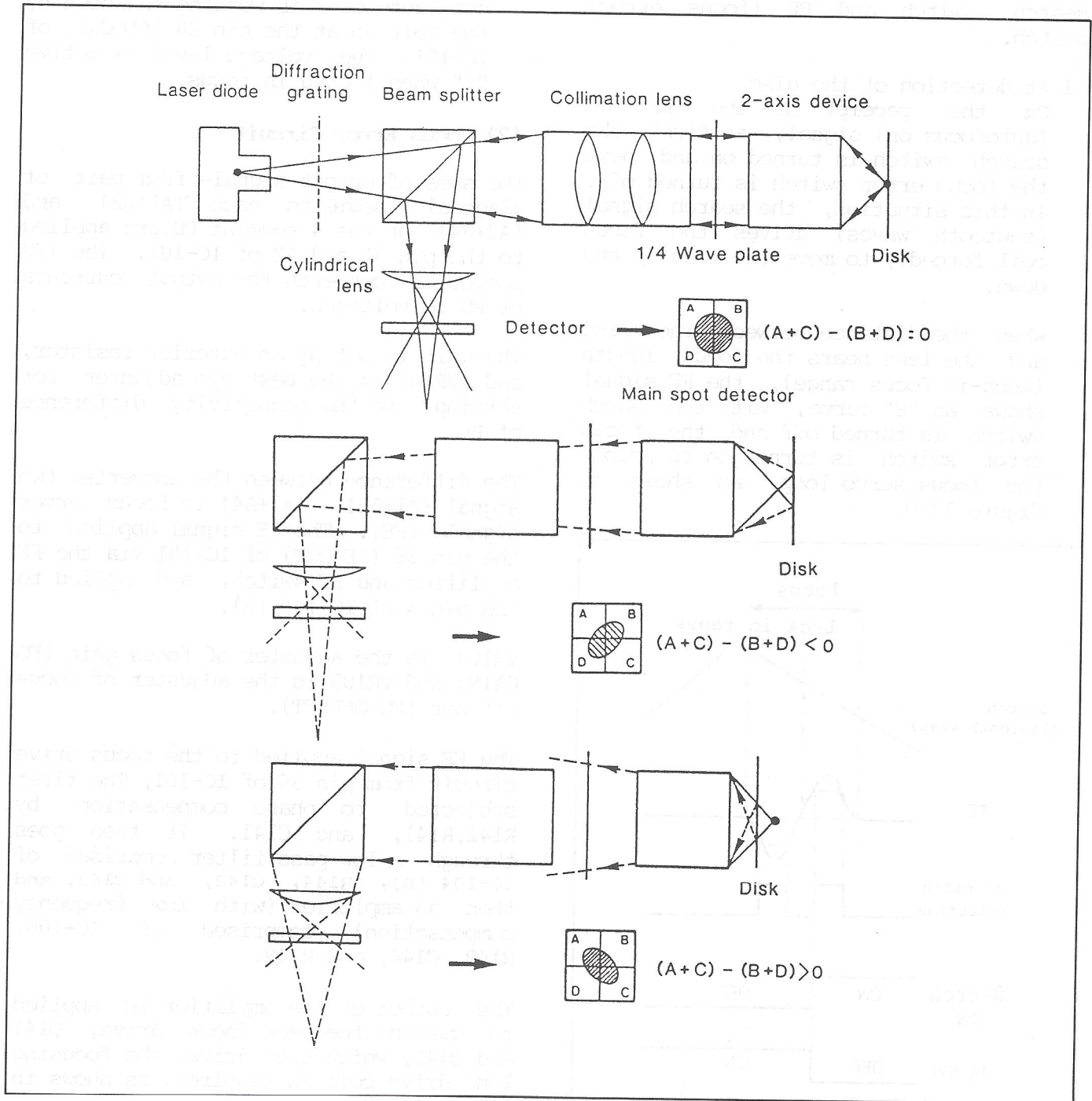


Figure 17-8 Detection of Focus Error

4) RF Amplifier and Dropout Detector Circuit

(1) RF Amplifier

The RF signal is a summation ($A1+A2+A3+A4+A5+A6$) of the output of all the 6 segment photodetector elements. Summation is performed within the optical pickup. The resulting RF signal is applied to

pin 33 of the servo IC-101, via transistor, where it is amplified by an RF amplifier having gain of approx, -34dB. The output of the RF amplifier appears at pin 31 of IC-101 and couples to the PLL/DATA SLICE circuit IC-501 pin 15.

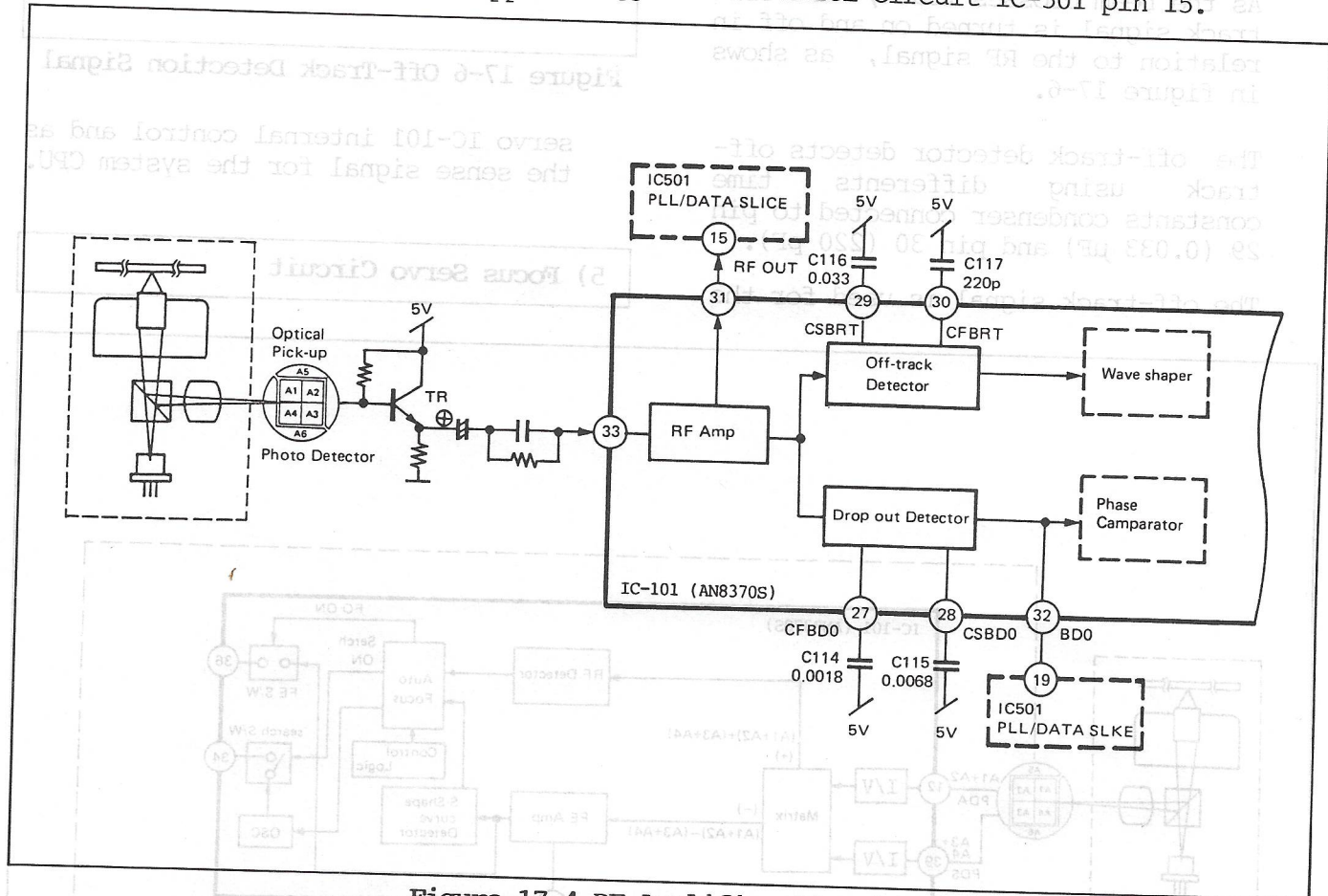


Figure 17-4 RF Amplifier Circuit

(2) Dropout Detector

The dropout detector detects signal dropouts in the RF signal, stabilizes the tracking servo against dropouts, and provides the detection signal to pin 32 of IC-101.

If a dropout occurs in the RF signal, the dropout detector detects it using different time constants condensers connected to pin 27 ($0.0018 \mu F$) and pin 28 ($0.0068 \mu F$). The dropout signal level is "H" to hold error signal before dropout. The dropout signal is applied to pin 32 of IC-101 to IC-501 pin 19.

The Process of detecting dropout is shown in figure 17-5.

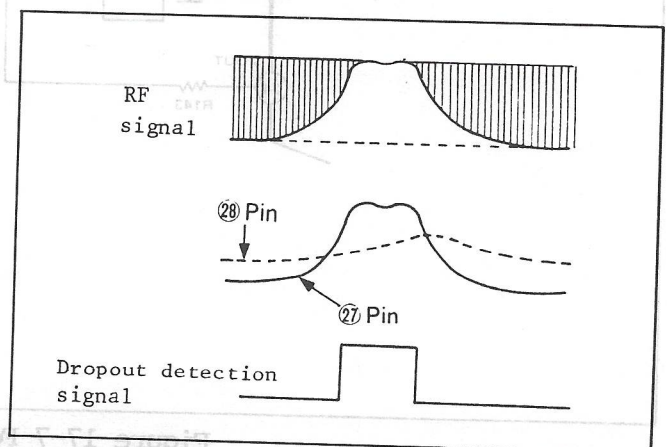


Figure 17-5 Dropout Detection Signal

(3) Off-Track Detector

The off-track detector detects the laser beam being off track. When tracking servo is turned off, the laser beam makes a traverse movement across the tracks during SEARCH, SKIP, mode even track-jump due to condition of the disc surface etc.

As the beam crosses track, the off-track signal is turned on and off in relation to the RF signal, as shows in figure 17-6.

The off-track detector detects off-track using different time constants condenser connected to pin 29 (0.033 μF) and pin 30 (220 pF).

The off-track signal is used for the

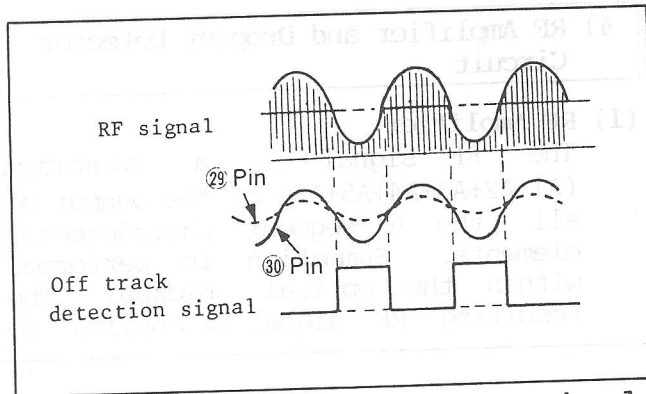


Figure 17-6 Off-Track Detection Signal

servo IC-101 internal control and as the sense signal for the system CPU.

5) Focus Servo Circuit

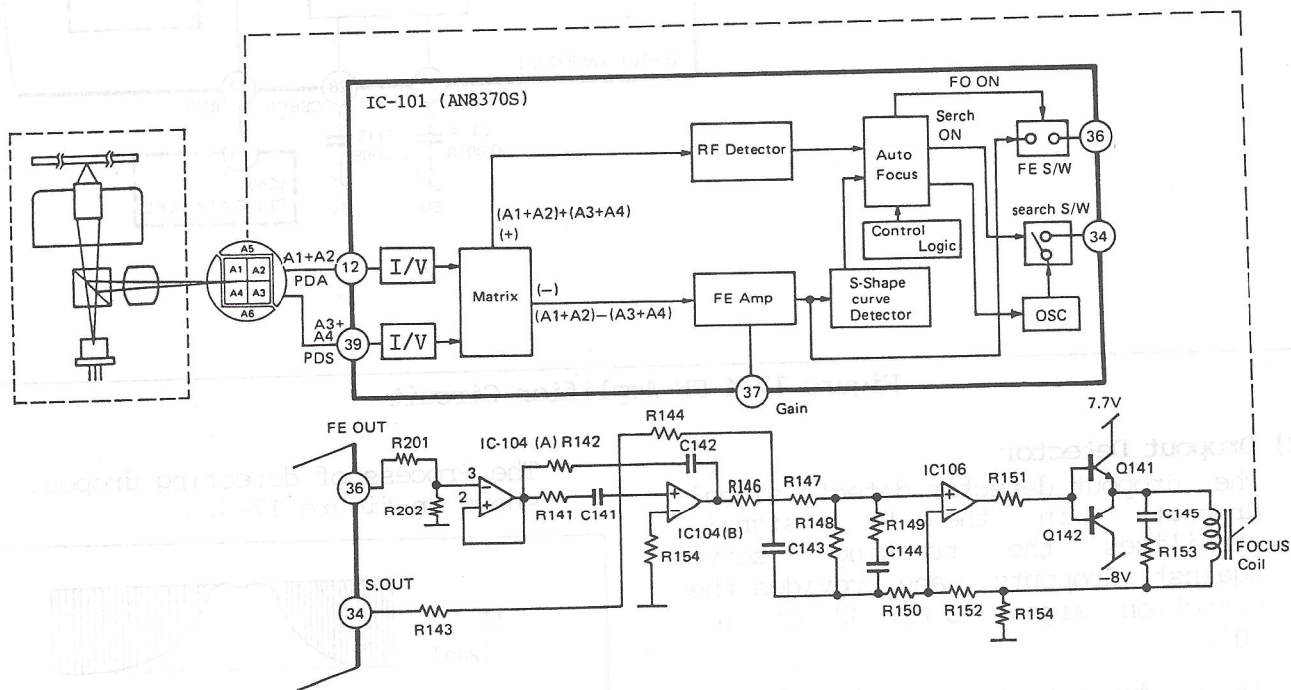


Figure 17-7 Focus Servo Circuit